

Hitesh Shimpi

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | hiteshshimpi45@gmail.com | +91-8055181811

Summary

Computer Science student skilled in Backend Development, Machine Learning, and Deep Learning, with hands-on experience building scalable APIs and AI-driven applications. Experienced with Python, FastAPI, REST APIs, AWS, data engineering, and machine learning, with a strong interest in backend systems, Docker-based containerization, and cloud infrastructure.

Education

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.72/10

Chennai, Tamil Nadu

Expected May 2027

KKJR Vidyalaya

Intermediate (Class XII)

Jalgaon, Maharashtra

Jun 2022 – Apr 2023

St. Joseph Convent School

Matriculation (Class X)

Jalgaon, Maharashtra

Jun 2020 – Apr 2021

Work Experience

Technical Intern (AICTE)

AI/ML Developer

Chennai, Tamil Nadu

Apr 2024 – Jun 2024

- Developed a deep learning-based rice leaf disease detection system using **EfficientNetB0** for 6-class image classification, with 224×224 image preprocessing, normalization, and data augmentation.
- Enhanced feature representation by integrating **CBAM attention** and **GeM pooling**, and trained the model using Adam optimizer and categorical cross-entropy with regularization to improve generalization and mitigate overfitting.

Projects

EduLink – Collaborative Study Platform

Jan 2025

- Developed a full-stack collaborative study platform supporting **real-time messaging, study groups, resource sharing, and user interactions**, with a scalable architecture designed to handle **20+ concurrent users**.
- Built backend services and **REST APIs** and optimized database queries, reducing application response latency by **30%** while integrating the frontend with backend services for seamless user interactions.
- Tech Stack: Python, FastAPI, REST APIs, React.js, Tailwind CSS, SQL, Git, GitHub*

Multi-Label Retinal Disease Classification – AI/ML

Feb 2026

- Engineered a deep learning-based multi-label retinal disease classification system using **EfficientNet and DenseNet** on **5,000+ retinal images** to classify multiple disease categories.
- Applied transfer learning for model development and used **Grad-CAM** visualizations to interpret model predictions and identify disease-relevant regions across multiple classes.
- Tech Stack: Python, TensorFlow, Keras, EfficientNet, DenseNet, Transfer Learning, Grad-CAM*

Skills Summary

Programming & Backend: Python, PostgreSQL, JavaScript, FastAPI, Node.js, REST APIs, Supabase

Web & Tools: React.js, Tailwind CSS, Next.js, Vue.js, Git, GitHub,

Data Engineering: Apache Spark, PySpark, Databricks, Lakeflow, ETL, Data Pipelines

Cloud & Machine Learning: AWS, TensorFlow, Keras, PyTorch, Scikit-learn, CNNs, EfficientNet, DenseNet

Soft Skills: Problem Solving, Team Collaboration, Communication

Certifications, Training & Patent

AWS Certified Cloud Practitioner

AWS Certified AI Practitioner

Databricks Badges: Apache Spark, Lakeflow, Data Pipelines, ETL

Patent - Smart Blind Spot Detection for Vehicles Using Multi-Sensor Integration and AI-Based Warning Mechanisms

Hackathons

- Guidewire DEVTRAILS '26**, 2026.
- Accenture Databricks**, 2026.
- Infosys HackWithInfy '26**, 2026.